

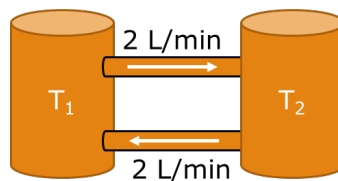
Chapter 4: System of Differential Equations

- 4.1. [System of Differential Equations](#) (P.1)
 - 4.2. [Solutions of System of Differential Equations](#) (P.6)
 - 4.3. [Phase Plane](#) (P.10)
 - 4.4. [Classification of Equilibrium Points](#) (P.16)
 - 4.5. [Real Distinct Eigenvalues](#) (P.25)
 - 4.6. [Complex Eigenvalues](#) (P.30)
 - 4.7. [Repeated Eigenvalues](#) (P.36)
-

4.1 System of Differential Equations

Systems of differential equations arise commonly from naturally occurring situations. We shall see how linear algebra concepts can be applied to analyse and solve certain system of differential equations. In this section, we will use two simple applications as examples to motivate the study of this topic. We will also give an overview of the different types of system of differential equations.

4.1.1 Mixing Problem Involving Two Tanks



Tank T₁ and T₂ contain initially 100 litre (L) of water each. In T₁ the water is pure, whereas 100kg of fertilizer are dissolved in T₂. By circulating liquid at a rate of 2 L/min and stirring (to keep the mixture uniform) the amounts of fertilizer in T₁ and T₂ change with time t. How much fertilizer will there be in T₁ and T₂ at time t (in min)?

Setting Up the Model

Amounts of fertilizer at time t: $y_1(t)$ for tank T₁, and $y_2(t)$ for tank T₂.

The (time) rate of change of $y_1(t)$ and $y_2(t)$ are given by $y_1'(t)$ and $y_2'(t)$ resp.

Our objective is to set up two equations to express $y_1'(t)$ and $y_2'(t)$ in terms of $y_1(t)$ and $y_2(t)$.

Amount of fertilizer flowing from T₁ to T₂ at time t is given by $\frac{2}{100} y_1(t)$

Amount of fertilizer flowing from T₂ to T₁ at time t is given by $\frac{2}{100} y_2(t)$

Since the rate of change in each tank = Inflow (of fertilizer)/min – Outflow (of fertilizer)/min

So we obtain

for tank T₁, $y_1'(t) = \frac{2}{100} y_2(t) - \frac{2}{100} y_1(t)$ and for tank T₂, $y_2'(t) = \frac{2}{100} y_1(t) - \frac{2}{100} y_2(t)$.

Hence we obtain a system of differential equations

$$\begin{aligned} y_1'(t) &= -0.02 y_1(t) + 0.02 y_2(t) \\ y_2'(t) &= 0.02 y_1(t) - 0.02 y_2(t) \end{aligned} \quad (*)$$

Furthermore, we are given that at the beginning, there is no fertilizer in tank 1 and 100kg of fertilizer in tank 2. This gives us the initial condition (at time $t = 0$):

$$y_1(0) = 0, y_2(0) = 100.$$

4.1.2 First Order Linear System of Differential Equations

The system (*) above is an example of a first order linear system of differential equations.

The term “first order” here refers to the highest derivatives involved in the equations being the first derivative $y_1'(t)$ and $y_2'(t)$.

The term “linear” has the same meaning as in our linear system, with $y_1(t)$ and $y_2(t)$ replacing the variables.

We say this is a 2×2 system since it has two variables $y_1(t)$ and $y_2(t)$ with two equations.

In this course, we shall use **SDE** to denote (first order linear) **S**ystem of **D**ifferential **E**quations.

We call this kind of system a coupled system since knowledge of $y_1(t)$ is required in order to solve for $y_2(t)$, and vice versa.

In the context of differential equations, it is understood that the objects $y_1(t)$ and $y_2(t)$ are functions, rather than numbers. Sometime we may not show the variable t by simply writing $y(t)$ as y , and $y'(t)$ as y' . In our example (*), we can do the same by hiding the t and write the system in a slightly simpler form:

$$\begin{aligned} y_1' &= -0.02 y_1 + 0.02 y_2 \\ y_2' &= 0.02 y_1 - 0.02 y_2 \end{aligned}$$

Since the system is “linear”, we can also write it in matrix form in the usual way:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -0.02 & 0.02 \\ 0.02 & -0.02 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

The column vector $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ is a vector function (i.e. a vector where the components are functions, instead of numbers). We denote it by $\mathbf{y}(t)$, or simply $\mathbf{y}$. Then the left hand side of the equation $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix}$ is the derivative of the vector function denoted by $\mathbf{y}'$. And we let the 2×2 matrix $\begin{pmatrix} -0.02 & 0.02 \\ 0.02 & -0.02 \end{pmatrix}$ be **A**. This matrix plays a crucial role in determining the solutions of the system (*).

So now we obtain a concise notation for the SDE:

$$\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t), \text{ or simply } \mathbf{y}' = \mathbf{A} \mathbf{y}.$$

and our objective is to solve for $\mathbf{y}(t)$, which will give us the two functions y_1 and y_2 .

4.1.3 Initial Value Problem

In the next section, we will show how to solve an SDE. There will be infinitely many solutions. In order to get a relevant particular solution, we need to specify an initial condition.

In our mixing problem, we are given $y_1(0) = 0$ and $y_2(0) = 100$.

The SDE together with the initial condition is known as an initial value problem.

first order linear SDE $\begin{aligned} y_1'(t) &= -0.02 y_1(t) + 0.02 y_2(t) \\ y_2'(t) &= 0.02 y_1(t) - 0.02 y_2(t) \end{aligned}$	Initial conditions: $\begin{aligned} y_1(0) &= 0 \\ y_2(0) &= 100 \end{aligned}$
 Initial Value Problem	

In matrix form, this is expressed as

$$\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t) \quad \mathbf{y}(0) = \begin{pmatrix} 0 \\ 100 \end{pmatrix}$$

4.1.4 Conversion of ODE to SDE

ODE stands for Ordinary Differential Equation, where the functions involved are single variable. (The term “ordinary” is used to distinguish from another class of partial differential equations that involved more than one variable in the functions.)

The example we have seen in our mixing problem consists of two ODE put together to form a system. Here is another example of ODE.

$$2y'' - 5y' + y = 0 \quad (\#)$$

You notice the equation involves the function y , together with its first and second derivatives y' and y'' . This is called a 2nd order ODE due to the second derivative term.

Let's also include the initial condition given by

$$y(3) = 6, \text{ and } y'(3) = -1 \quad (\Delta)$$

For a 2nd order ODE, we need an initial value for both y and y' .

As the main focus of this course is on system of differential equations, what we will do is to convert this 2nd order ODE (#) to a (first order) 2 x 2 SDE. This can be done with a very simple change of variable:

Define the following two new functions y_1 and y_2 as follow:

$$y_1(t) = y(t) \quad (1)$$

$$y_2(t) = y'(t) \quad (2)$$

In other words, y_1 is simply the function y in the ODE (#), and y_2 is its first derivative y' .

If we differentiate both sides of (1) and (2), we get

$$y_1'(t) = y'(t) \quad (1')$$

$$y_2'(t) = y''(t) \quad (2')$$

In order to convert the ODE to an SDE, we want (1') and (2') to be expressed in terms of y_1 and y_2 .

For the first equation, y' can be substituted by y_2 in view of (2), and for the second equation, we use the original ODE (#) to derive

$$y'' = -\frac{1}{2} y + \frac{5}{2} y' = -\frac{1}{2} y_1 + \frac{5}{2} y_2$$

where the second equality comes from (1) and (2).

We can also convert the above initial condition (Δ) into $y_1(3) = 6$ and $y_2(3) = -1$.

Putting all these together, we have converted a 2nd order ODE into a first order SDE.

$2y'' - 5y' + y = 0$ $y(3) = 6 \text{ and } y'(3) = -1$	$\longleftrightarrow$	$\begin{aligned} y_1' &= y_2 & y_1(3) &= 6 \\ y_2' &= -\frac{1}{2}y_1 + \frac{5}{2}y_2 & y_2(3) &= -1 \end{aligned}$
2 nd order ODE		first order linear SDE

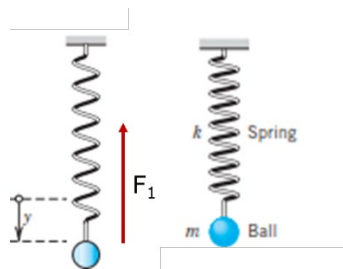
We can put it in matrix form:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{5}{2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad \mathbf{y}(3) = \begin{pmatrix} y_1(3) \\ y_2(3) \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$$

4.1.5 Mass–Spring System

This is another classical example of modeling the free motions of a mass on a spring, using a 2nd order ODE.

We take an ordinary coil spring that resists extension as well as compression. Suspend it vertically from a fixed support and attach an iron ball at its lower end.



We now pull the ball down by a distance y as shown. In other words, y is the displacement of the ball from rest, and we take the downward direction as positive.

Now the spring will exert a spring force F_1 . By Hooke's law, this force is proportional to the stretch y , with the spring constant k (> 0):

$$F_1 = -ky \quad (\text{i})$$

The minus sign indicates that the force is against the displacement. It is a restoring force: It wants to restore the system, that is, to pull the ball back to $y = 0$.

Every system has damping. Otherwise it would keep moving forever. But if the damping is small and the motion of the system is considered over a relatively short time, we may disregard damping.

If we assume the system is undamped, then Newton's law gives the model:

$$my'' = F_1 \quad (\text{ii})$$

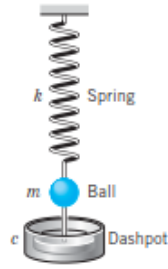
Combining (i) and (ii) gives this 2nd order ODE:

$$y'' = -\frac{k}{m}y$$

which describe a motion known as harmonic oscillation.

4.1.6 Damped Mass – Spring System

We now connect the ball to a dashpot as shown to add a damping force F_2 .



This damping force is given to be proportional to the velocity y' , with the damping constant c .

$$F_2 = -cy' \quad (\text{iii})$$

When the motion is downward, F_2 is an upward force; when the motion is upward, F_2 becomes a downward force. So the Newton's Second law gives

$$my'' = F_1 + F_2 = -ky - cy'$$

by using (i) and (iii).

So now we obtained the ODE of the damped mass–spring system:

$$m y'' + c y' + ky = 0.$$

By substituting y and y' with y_1 and y_2 , we can convert this ODE into a 2×2 SDE

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= -\frac{k}{m} y_1 - \frac{c}{m} y_2 \end{aligned}$$

The matrix form of this SDE is given below:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

4.2 Solutions of System of Differential Equations

In this section, we will see how to solve an SDE. You will see the application of the concepts of eigenvalues and eigenvectors in the process.

4.2.1 Solutions in terms of eigenvalues and eigenvectors

Recall that an SDE can be written in this matrix form

$$\mathbf{y}' = \mathbf{A}\mathbf{y} \quad (1)$$

where $\mathbf{A}$ is a 2×2 matrix. Suppose $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with corresponding eigenvalue r . i.e.

$$\mathbf{A}\mathbf{v} = r\mathbf{v}.$$

Then the vector function $\mathbf{y}(t)$ given by

$$\mathbf{y}(t) = \mathbf{v} e^{rt} \quad (2)$$

where e^{rt} is the exponential function in variable t , is a solution of the SDE (1).

To see this, we first differentiate (2) with respect to t to get

$$\mathbf{y}'(t) = r\mathbf{v} e^{rt} \quad (3)$$

Note that the vector $\mathbf{v}$ is independent of t . So we only need to differentiate the exponential function.

On the other hand, substituting (2) to the right hand side of (1):

$$\mathbf{A}\mathbf{y} = \mathbf{A}\mathbf{v} e^{rt} = r\mathbf{v} e^{rt} \quad (4)$$

Since the right hand sides of both (3) and (4) are the same, we get $\mathbf{y}' = \mathbf{A}\mathbf{y}$, i.e. the SDE (1) is satisfied by (2).

Therefore, to solve the SDE (1), we must first find the eigenvalues and eigenvectors of $\mathbf{A}$.

4.2.2 Solving Linear SDE

$$\begin{cases} y_1' &= -3y_1 + y_2 \\ y_2' &= y_1 - 3y_2 \end{cases}$$

This is a 2×2 SDE with matrix form $\mathbf{y}' = \mathbf{A}\mathbf{y}$ where $\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$.

Let's find the eigenvalues r of $\mathbf{A}$ by solving the characteristic polynomial:

$$\det(\lambda\mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda+3 & -1 \\ -1 & \lambda+3 \end{vmatrix} = \lambda^2 + 6\lambda + 8$$

This gives the eigenvalues $r_1 = -2$ and $r_2 = -4$.

It is not difficult to solve for the eigenvectors: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2 .

So each eigen-pair will give us a solution of the system:

$$\mathbf{y}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} \quad \text{and} \quad \mathbf{y}(t) = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-4t} = \begin{pmatrix} -e^{-4t} \\ e^{-4t} \end{pmatrix}.$$

We can also write the solutions in terms of the component functions:

$$y_1 = e^{-2t}, y_2 = e^{-2t} \quad \text{and} \quad y_1 = -e^{-4t}, y_2 = e^{-4t}.$$

These are not the only two solutions. The Superposition principle tells us that there are infinitely many solutions.

4.2.3 Superposition Principle

This principle states that,

If $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are two solutions to an SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$, then

$$\mathbf{y} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) \quad (*)$$

is also a solution to the system for any scalars c_1 and c_2 .

In the previous example, we have found two solutions for our system:

$$\begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} \text{ and } \begin{pmatrix} -e^{-4t} \\ e^{-4t} \end{pmatrix}.$$

This allows us to generate infinitely many solutions of the form given by this linear combination:

so $c_1 \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-4t} \\ e^{-4t} \end{pmatrix}$ is also a solution of the system.

To see why the Superposition principle works, we just need to show the linear combination (*) satisfies the system.

First we differentiate (*) to get

$$\mathbf{y}' = c_1\mathbf{x}_1'(t) + c_2\mathbf{x}_2'(t)$$

Since $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions of the system, we can replace $\mathbf{x}_1'$ and $\mathbf{x}_2'$ with $\mathbf{A}\mathbf{x}_1$ and $\mathbf{A}\mathbf{x}_2$ in the equation:

$$\mathbf{y}' = c_1\mathbf{A}\mathbf{x}_1(t) + c_2\mathbf{A}\mathbf{x}_2(t)$$

Now by property of matrix multiplication, we can rewrite the right hand side to get

$$\mathbf{y}' = \mathbf{A}(c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t))$$

But the term in the bracket on the right hand side is just $\mathbf{y}$ from (*).

Hence we have shown that the linear combination (*) is a solution of the system $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

4.2.4 Fundamental Set of Solutions

Let $\mathbf{y}' = \mathbf{A}\mathbf{y}$ be a 2×2 SDE. Suppose $\mathbf{v}_1$ and $\mathbf{v}_2$ are two eigenvectors of $\mathbf{A}$ with corresponding eigenvalues r_1 and r_2 (note that r_1 and r_2 may be equal). Then $\mathbf{y} = \mathbf{v}_1 e^{r_1 t}$ and $\mathbf{y} = \mathbf{v}_2 e^{r_2 t}$ are two solutions of the SDE, and by Superposition principle,

$$\mathbf{y} = c_1\mathbf{v}_1 e^{r_1 t} + c_2\mathbf{v}_2 e^{r_2 t} \quad (\Delta)$$

is also a solution of the SDE for any scalar c_1 and c_2 .

Is it possible to find other solutions of the SDE which is not given by (Δ)?

If $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent eigenvectors, then (Δ) will generate all the solutions of the SDE.

In this case, we say (Δ) is a general solution of the SDE.

The two solutions

$$\mathbf{y} = \mathbf{v}_1 e^{r_1 t} \text{ and } \mathbf{y} = \mathbf{v}_2 e^{r_2 t} \quad (\#)$$

are not “scalar multiple” of each other, and they are called a fundamental set of solutions of the SDE.

On the other hand, if $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly dependent eigenvectors, then (Δ) will not generate all the solutions of the SDE. In this case, the two solutions in $(\#)$ are “scalar multiple” (see below*) of each other, and so they do not form a fundamental set of solutions of the SDE.

(*Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly dependent, $\mathbf{v}_2 = k\mathbf{v}_1$. So the second solution $\mathbf{v}_2 e^{r_2 t} = k\mathbf{v}_1 e^{r_2 t} = ke^{(r_2-r_1)t} \mathbf{v}_1 e^{r_1 t}$, which is a scalar multiple of the first solution.)

Example:

In our earlier example

$$\begin{cases} y_1' &= -3y_1 + y_2 \\ y_2' &= y_1 - 3y_2 \end{cases}$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ has two linearly independent eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

So the two solutions:

$$\begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} \text{ and } \begin{pmatrix} -e^{-4t} \\ e^{-4t} \end{pmatrix}$$

form a fundamental set of solutions of our SDE, and

$$\mathbf{y} = c_1 \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-4t} \\ e^{-4t} \end{pmatrix} \text{ is the general solution of the system.}$$

i.e. all solution of the system is given by the linear combination of these two solutions.

4.2.5 Existence and Uniqueness of Solutions

Given an SDE, we can find infinitely many solutions.

If we have an initial value problem:

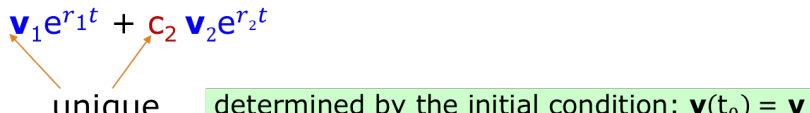
$$\mathbf{y}'(t) = \mathbf{A} \mathbf{y}(t)$$

$$\mathbf{y}(t_0) = \mathbf{v}$$

then the solution will be unique. In other words, there will be exactly one solution that can satisfy the system and the given initial condition.

More specifically, if the general solution of the system is given by this linear combination,

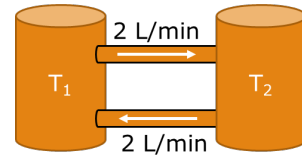
$$\mathbf{y} = c_1 \mathbf{v}_1 e^{r_1 t} + c_2 \mathbf{v}_2 e^{r_2 t}$$



then the initial value problem will have a solution given by exactly one pair of values of c_1 and c_2 , determined by the initial condition.

4.2.6 Mixing Problem Involving Two Tanks

Let's go back to our mixing tank problem from the previous section.



Recall that the model is given by the SDE with the underlying 2×2 matrix $\mathbf{A} = \begin{pmatrix} -0.02 & 0.02 \\ 0.02 & -0.02 \end{pmatrix}$.

It is easy to check that this matrix has

eigenvalues $r_1 = 0$ and $r_2 = -0.04$, and eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2 .

Since the two eigenvectors are linearly independent, we have a fundamental set of solutions

$$\mathbf{y} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{0t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-0.04t} = \begin{pmatrix} -e^{-0.04t} \\ e^{-0.04t} \end{pmatrix}.$$

Note that the first solution is in fact independent of t due to the 0 eigenvalue.

Now the linear combination of these two solutions gives the general solution of the system:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -e^{-0.04t} \\ e^{-0.04t} \end{pmatrix}.$$

With the given initial condition: $y_1(0) = 0$, $y_2(0) = 100$, we proceed to solve the initial value problem to get the unique solution:

In the general solution above, we substitute $t = 0$ to the linear combination to get this next equation:

$$\mathbf{y}(0) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 100 \end{pmatrix}$$

The right hand side of the equation come from the initial condition.

Note that this is in fact a regular 2×2 linear system, and we can solve for c_1 and c_2 easily. We will get the specific values $c_1 = 50$ and $c_2 = 50$. So we have this particular linear combination

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 50 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 50 \begin{pmatrix} -e^{-0.04t} \\ e^{-0.04t} \end{pmatrix}$$

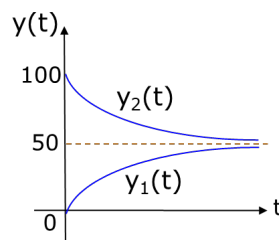
as the unique solution of the initial value problem.

This gives us the two functions

$$y_1(t) = 50(1 - e^{-0.04t}) \quad \text{and} \quad y_2(t) = 50(1 + e^{-0.04t})$$

that represents the amount of fertilizers in the two tanks at time t .

Let's sketch the graphs of these two functions.



As the exponential function is raised to a negative power, when t increases, the exponential function decreases. So we see that y_1 increases over time from 0 to 50kg, while y_2 decreases from 100 to 50kg. Therefore, in the long run, the amount of fertilizers in both tanks will be about the same.

4.3. Phase Plane

In this section, we will show how to plot the graphs of the solutions of 2 x 2 SDE.

4.3.1 Graphing Vector Function

For a general vector function with two components $\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$

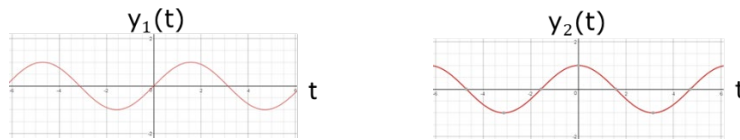
There are two ways to plot its graph:

- two curves over the t -axis, one for each component $y_1(t)$ and $y_2(t)$;
- a single curve in the $y_1 y_2$ -plane.

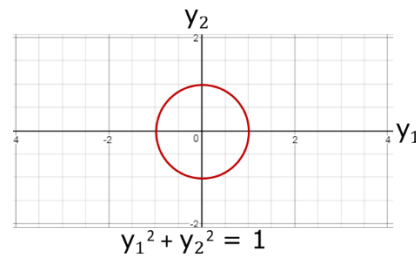
Example:

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = \begin{pmatrix} \sin(t) \\ \cos(t) \end{pmatrix}$$

Separately, the graphs of the two trigonometric functions are given by:



Alternatively, we can draw the parametric curve on the y_1 - y_2 plane, which is the unit circle.



To obtain a single equation of y_1 and y_2 , we try to eliminate the variable t from the two functions. For this particular vector function, we observe that

$$y_1^2 + y_2^2 = \sin^2(t) + \cos^2(t) = 1$$

which gives us $y_1^2 + y_2^2 = 1$, the equation of the unit circle.

However, in general, it is not easy to come up with the equation in such a way.

4.3.2 Graphing Solutions of SDE

Consider the SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$:

$$\begin{cases} y_1' &= y_1 + 2y_2 \\ y_2' &= 3y_1 + 2y_2 \end{cases}$$

with underlying matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.

This gives the eigenvalues $r_1 = 4$ and $r_2 = -1$ with eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2 (check!).

So the general solution of the SDE is given by

$$\mathbf{y} = c_1 \begin{pmatrix} 2e^{4t} \\ 3e^{4t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-t} \\ e^{-t} \end{pmatrix} = \begin{pmatrix} 2c_1e^{4t} - c_2e^{-t} \\ 3c_1e^{4t} + c_2e^{-t} \end{pmatrix}.$$

In terms of the two components y_1 and y_2 , we have:

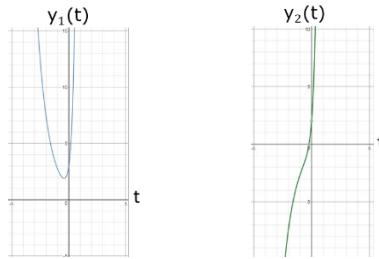
$$y_1 = 2c_1e^{4t} - c_2e^{-t} \quad \text{and} \quad y_2 = 3c_1e^{4t} + c_2e^{-t}$$

We need to specify the values of c_1 and c_2 in order to plot the graph.

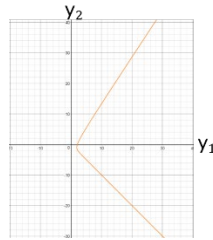
Example: If we take $c_1 = 1, c_2 = -1$, then

$$y_1 = 2e^{4t} + e^{-t} \quad \text{and} \quad y_2 = 3e^{4t} - e^{-t}$$

Then separately, the graphs of y_1 and y_2 against the t -axis looks like these:



On the other hand, the plot on the y_1 - y_2 plane will be given by this graph:



There are software to assist in such graph plotting (e.g. Desmos - parametric graphing: <https://www.desmos.com/calculator/ksjcpazwa9>). In this course, we will learn how to use eigenvalue and eigenvector to give a qualitative description of the graph, which tells us the behaviour of the solutions of the system.

In the study of SDE, the y_1 - y_2 plane is called the phase plane, as the underlying variable t usually represent the time. The graph of the solution in the phase plane is known as a trajectory of the system, suggesting that there is a direction of the curve when the value of t increase from negative infinity to 0 to positive infinity. We will make this clearer in the subsequent discussion.

4.3.3 Phase Portrait

What we have shown above is the trajectory corresponding to only one of the solutions of the SDE

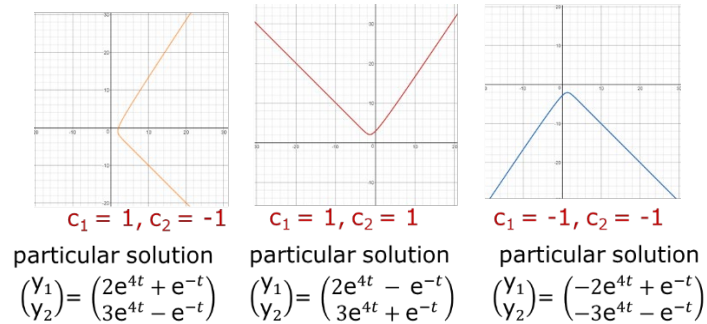
$$\begin{cases} y_1' &= y_1 + 2y_2 \\ y_2' &= 3y_1 + 2y_2 \end{cases}$$

with general solution

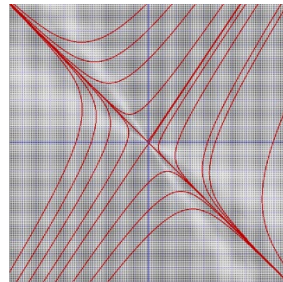
$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2c_1e^{4t} - c_2e^{-t} \\ 3c_1e^{4t} + c_2e^{-t} \end{pmatrix}$$

Every pair of c_1 and c_2 will give a particular solution, and hence a different trajectory.

Here are a few more particular solutions of the system and their respective trajectories.



By repeating this with different pairs of c_1 and c_2 , we can trace many trajectories of the system. If we fill the phase plane with a few representative trajectories, we obtain the phase portrait of the system.

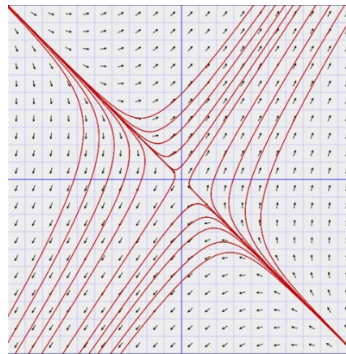


phase portrait of the system

Studies of solutions in the phase plane have become quite important along with advances in computer graphics, because a phase portrait gives a good general qualitative impression of the entire family of solutions, which we shall elaborate in subsequent sections.

4.3.4 Direction Field

Sometime we see tiny arrows being drawn on the phase plane, as shown below:

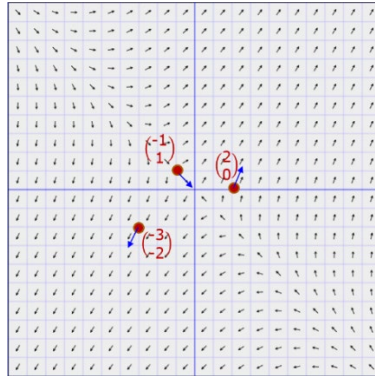


These arrows indicate the directions of the tangent vectors to the trajectories at the various points on the plane. We refer to them collectively as the direction field of the SDE.

In an SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$, the values of $\mathbf{y}$ correspond to points in the phase plane, and the values of $\mathbf{y}'$ represents the tangent to the trajectory at the point representing $\mathbf{y}$. And this tangent can be obtained from the point by pre-multiplying $\mathbf{A}$ to $\mathbf{y}$.

Let's illustrate with our example $\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{y}$.

If we take (randomly) $\mathbf{y} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ (red dot), then $\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (blue arrow). This means that the trajectory at the point $(-1, 1)$ on the phase plane will be moving in the direction of the tangent $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.



Note that only the direction of the tangent vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is relevant here. So we just need to draw a short arrow as shown.

Similarly, if we take $\mathbf{y} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, then $\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$. So we draw an arrow starting from the point $(2, 0)$ and in the direction of the tangent $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$.

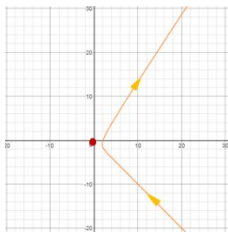
And if we let $\mathbf{y} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}$, then we get the tangent vector $\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} -3 \\ -2 \end{pmatrix} = \begin{pmatrix} -7 \\ -13 \end{pmatrix}$ at the point $(-3, -2)$.

The direction of the arrows collectively will tell us the direction of each trajectory as t increases. This helps us to visualize the time dependence of the solution. There are some web applets on the internet that plot the trajectories of a given system with direction field. For example:

<http://www.bluffton.edu/homepages/facstaff/nesterd/java/slopefields.html>

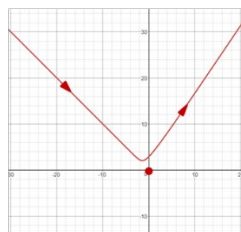
4.3.5 Behavior of Trajectories (An Example)

The following are four trajectories of our system $\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{y}$ (corresponding to 4 particular solutions). We put in arrows on the trajectories to indicate the directions of the trajectories as t increases. (These directions tally with the small arrows on the direction field.)



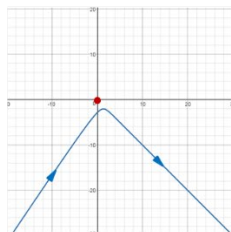
$$c_1 = 1, c_2 = -1$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2e^{4t} + e^{-t} \\ 3e^{4t} - e^{-t} \end{pmatrix}$$



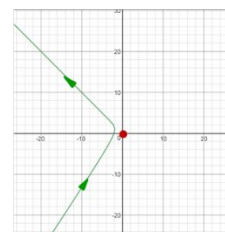
$$c_1 = 1, c_2 = 1$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2e^{4t} - e^{-t} \\ 3e^{4t} + e^{-t} \end{pmatrix}$$



$$c_1 = -1, c_2 = 1$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -2e^{4t} - e^{-t} \\ -3e^{4t} + e^{-t} \end{pmatrix}$$



$$c_1 = -1, c_2 = -1$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -2e^{4t} + e^{-t} \\ -3e^{4t} - e^{-t} \end{pmatrix}$$

It looks like most of the solutions will start away from the origin, then as t starts to increase they move in towards the origin, and then eventually start moving away from the origin again.

This can be explained by the behavior of the exponential functions appearing in the solutions.

As t increases, the term e^{4t} increases from 0 to infinity, since $4t$ is a positive power.

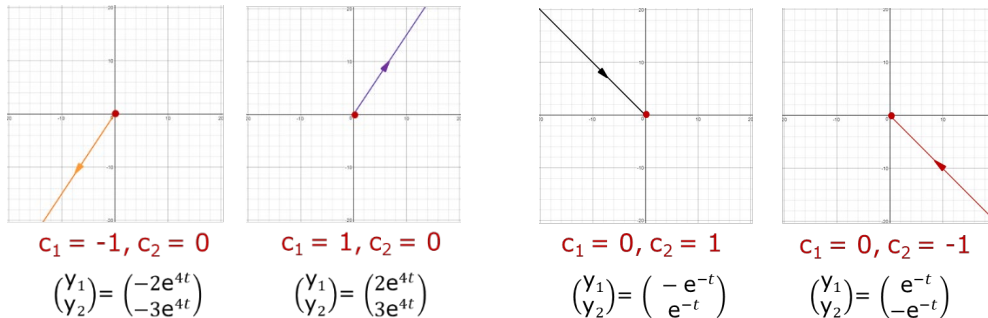
On the other hand, the term e^{-t} decreases from infinity to 0, since $-t$ is a negative power.

For small (very negative) t , e^{-t} will dominate e^{4t} , so the trajectory will start far away from the origin.

For large (very positive) t , e^{4t} will dominate e^{-t} , so the trajectory will move away from the origin in the long run.

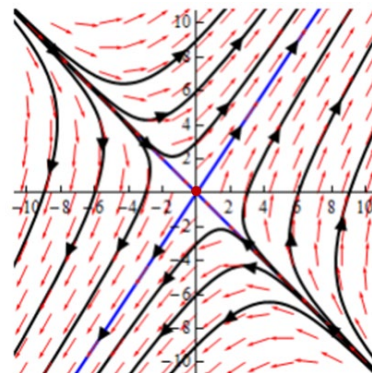
There are four solutions of the system that have slightly different behaviors.

Two of the solutions with $c_2 = 0$ will only have the e^{4t} terms in the components. The trajectories will start near the origin and then move straight away from it. (See diagrams on the left.)



The other two solutions with $c_1=0$ will only have e^{-t} terms in the components. Their trajectories start away from the origin and then move straight towards it. (See diagrams on the right.)

The following diagram shows the phase portrait together with the directions of the trajectories.



We shall see in later sections that, for different SDE, the trajectories may behave very differently from the example we have seen above.

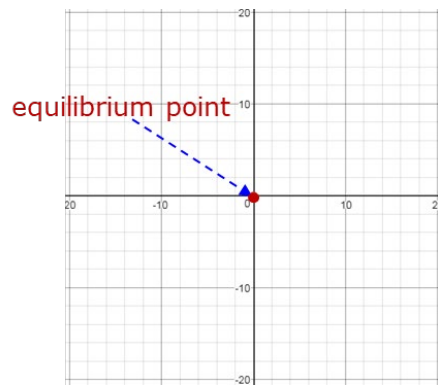
4.3.6 Equilibrium Solution

Given any SDE: $\mathbf{y}' = \mathbf{A}\mathbf{y}$, one obvious solution is

$$\mathbf{y} = \mathbf{0}$$

It is independent of t , and is called an equilibrium solution for the system. In most cases, $\mathbf{y} = \mathbf{0}$ is the only equilibrium solution. But in some exceptional cases, there can be infinitely many such solutions that are independent of time.

The graph of this solution is not a trajectory, but instead it is represented by a point, corresponding to the origin of the phase plane. We call it an equilibrium point (or critical point).



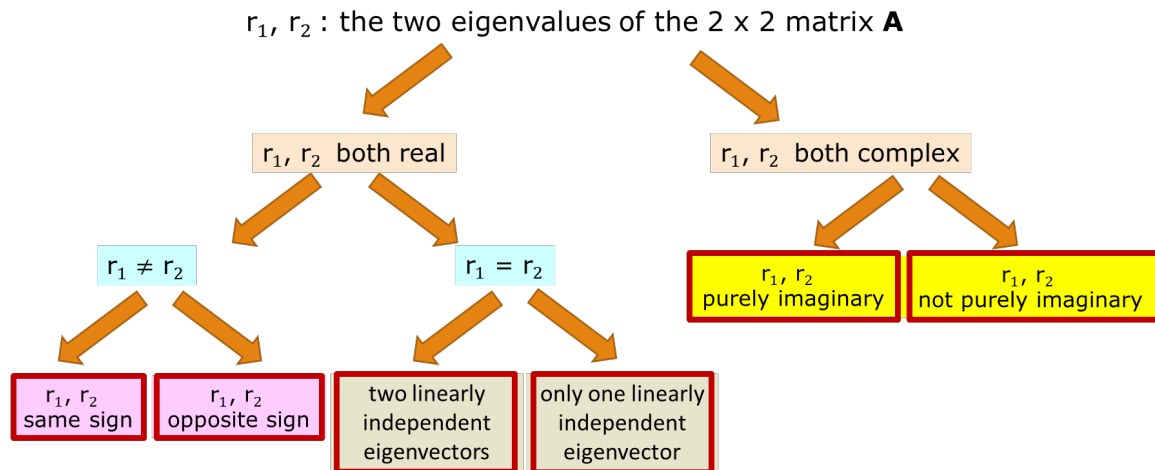
In the next section, we are going to classify equilibrium points of SDE based on the behavior of the trajectories around them.

4.4. Classification of Equilibrium Points

In this section, we take a closer look at the different types of equilibrium points of a 2×2 SDE. We shall see how the eigenvalues can help us to classify the equilibrium points.

4.4.1 Eigenvalues of 2×2 Matrices

The following diagram lists all possible scenarios for the two eigenvalues of a 2×2 matrix.



When the eigenvalues are complex numbers, they must come in conjugate pairs, namely, if $r_1 = a + bi$, then $r_2 = a - bi$. Also recall a complex number is purely imaginary if the real part $a = 0$. The six scenarios boxed up in red correspond to the six main types of equilibrium points in SDE.

4.4.2 Types of Equilibrium Points

The six types of equilibrium points are given in the table (right column) below

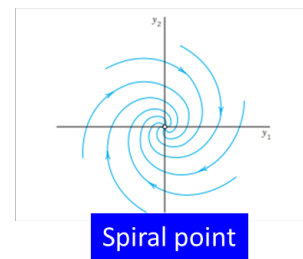
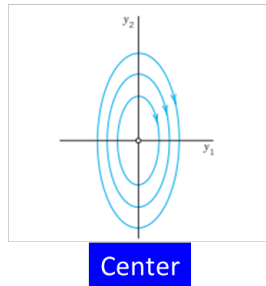
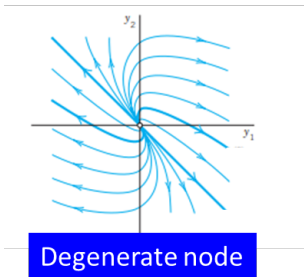
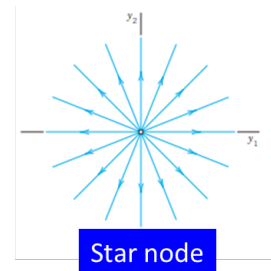
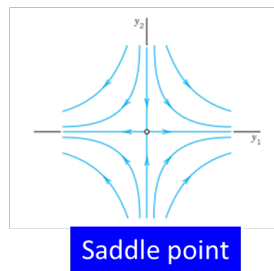
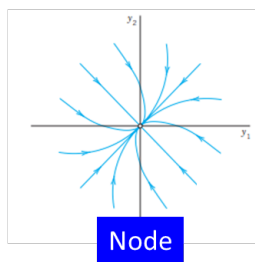
Let r_1, r_2 be the two eigenvalues (can be repeated or complex) of the underlying 2×2 matrix **A** of the SDE. The left column are the different scenarios on the eigenvalues of **A** discussed above.

Eigenvalues r_1 and r_2	Equilibrium Point
Real, distinct, same sign	Node
Real, distinct, opposite sign	Saddle point
Real, repeated, 2 lin. indep. eigenvectors	Star node
Real, repeated, only 1 lin. indep. eigenvector	Degenerate node
Pure imaginary	Center
Complex but not pure imaginary	Spiral point

In the subsequent discussion, we shall see each type of equilibrium points is characterized by its phase portrait.

4.4.3 Phase Portraits

Below are the phase portraits that correspond to the six types of equilibrium points.



Let's give some informal description of the phase portrait for the different types of equilibrium point. For a node, the main feature is that all the trajectories "converge" to the equilibrium at the origin, and are somewhat symmetrical about the two axes.

For a saddle point, except for two pairs of straight trajectories, all other trajectories are shaped like saddles.

For a star node, all trajectories "converge" to the origin like the node, but in this case, every trajectory is a straight line.

For a degenerate node, while all trajectories "converge" to the origin, unlike the node, whose phase portrait is "symmetrical" about the axes, the degenerate node is asymmetrical.

For a center, the trajectories are a bunch of close loops centered at the origin.

For a spiral point, all trajectories spiral around the origin.

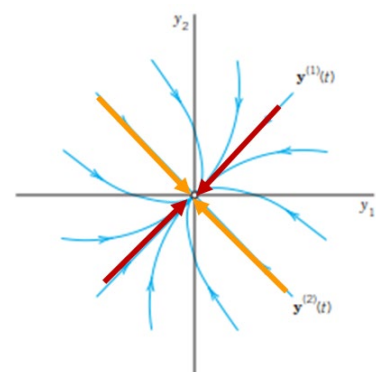
We shall give a more detailed description of the behavior for each type of phase portraits below.

One thing to take note is the arrows on the trajectories that are drawn on the phase portrait. We shall see later that these arrows can be pointing in one direction or the other uniformly.

4.4.4 Node

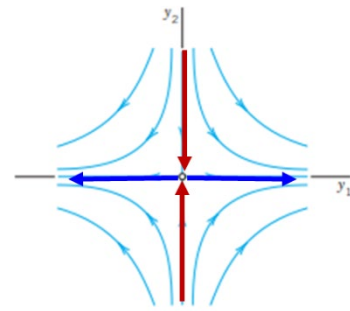
A node has two distinct eigenvalues, and hence two linearly independent eigenvectors. These two eigenvectors determine two axes for the phase portrait. Along one of the axes, there are two straight line with one end converging to the origin. Similarly, along the other axis, there are also two straight line trajectories with one end converging to the origin. All other trajectories are curves, and have the same limiting direction at the origin with one of the axes.

We see here that, the directions of the trajectories all moving towards the origin. It is also possible for all the trajectories to move away from the origin (in the opposite direction).



4.4.5 Saddle Point

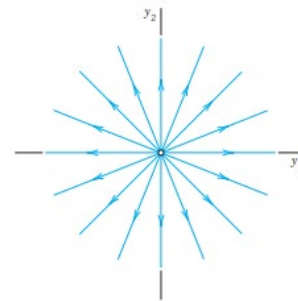
A saddle point also has two distinct eigenvalues, and hence two linearly independent eigenvectors. Again, these two eigenvectors determine two axes for the phase portrait. Along one axis, there are two straight-line trajectories with one end “converging” to the origin. And along the other axis, there are also two straight-line trajectories with one end “converging” to the origin. The difference here is, one pair of the straight-line trajectories moves towards the origin, and the other pair moves away from it. All other trajectories are curved, start parallel to one axis and end parallel to the other axis, bypassing the origin.



4.4.6 Star Node

First of all, we make a remark that all star nodes come from 2×2 scalar matrices. And for such a matrix, though it has only one eigenvalue, every nonzero 2-vector is an eigenvector. This is why the phase portrait of a star node consists of only straight-line trajectories with one end “converging” to the origin. There is a trajectory in every direction, each corresponding to a non-zero 2-vector.

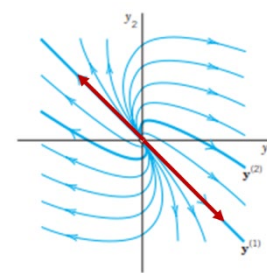
We see here that, the directions of the trajectories all moving away from the origin. It is also possible for all the trajectories to move towards the origin.



4.4.7 Degenerate Node

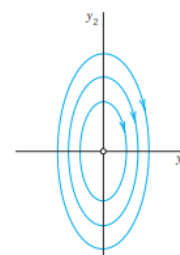
A degenerate node has only one linearly independent eigenvector. All other eigenvectors are scalar multiples of the same vector. Therefore, the phase portrait has only one “axis” determined by this eigenvector. Like the node and star node, all trajectories of degenerate node have one end converging to the origin. Along the axis, there are two straight line trajectories. All other trajectories are on the two sides of the axis. They all start parallel to the axis, make a turn around, and end parallel to the same axis again. The trajectories on the two sides of the axis moves in “opposite ways”.

We see here that, the directions of the trajectories all moving away from the origin. It is also possible for all the trajectories to move towards the origin.



4.4.8 Center

Centers come from eigenvalues that are complex and purely imaginary. Since the eigenvectors are also complex, there is no axis on the phase portrait that is determined by the eigenvector. Instead, the phase portrait consists of infinitely many concentric closed loops enclosing the origin. In a later section, it will become



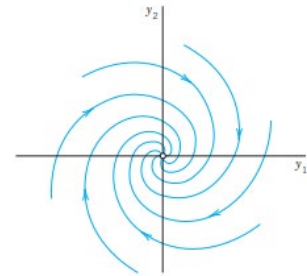
clearer the behavior of the trajectories for a center when we convert the complex solution of an SDE into a real solution.

We make a remark about the shape of the loops: they can be circles or ellipses; they can be upright or tilted. (In this course, students are not expected to sketch the exact shape of the trajectories.)

We see here that, the orientation of the loops are all clockwise, but it is also possible for the orientation to be all counterclockwise.

4.4.9 Spiral Point

Like the centers, the eigenvalues and eigenvectors are complex, so there is no axis on the phase portrait that is determined by the eigenvectors. Instead, the phase portrait consists of trajectories that spiral around the origin, with one end converging to it. This behavior will again be explained in a later section when we analyse the real solution of the SDE.



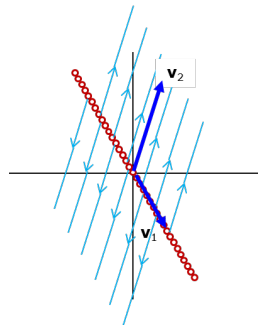
Some remark about the orientation and direction of the spirals:

We see here that the orientation of the spirals are all clockwise, but it is also possible for the orientation to be all counterclockwise. We also see here that the direction of the spirals are all moving towards the origin, or but is also possible for the spirals to be all moving away from it.

4.4.10 Zero Eigenvalue

This is an exceptional case with infinitely many equilibrium points.

If one of the eigenvalues of $\mathbf{A}$ is 0 with eigenvector $\mathbf{v}_1$, while the other eigenvalue is nonzero with eigenvector $\mathbf{v}_2$. All the points (in red) along the line parallel to $\mathbf{v}_1$ are equilibrium points, while all trajectories are straight lines parallel to $\mathbf{v}_2$.



Consider this SDE

$$\begin{cases} y_1' &= 2y_1 + y_2 \\ y_2' &= 6y_1 + 3y_2 \end{cases} \quad \text{with matrix form } \mathbf{y}' = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix} \mathbf{y}.$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}$ has eigenvalues $r_1 = 0$ and $r_2 = 5$ with eigenvectors

$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ for r_2 .

[Note: if we solve for the eigenvectors using Gaussian elimination, we should get

$\mathbf{v}_1 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1/3 \\ 1 \end{pmatrix}$. Since any scalar multiples of eigenvectors will still be eigenvectors, we

may use $-2\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and $3\mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, which have integer entries, to replace the original ones.]

In this case, we can still get two solutions of the system.

$$\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ and } \mathbf{x}_2(t) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{5t}.$$

The solution $\mathbf{x}_1$ corresponding to 0 eigenvalue will be a constant vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, instead of a vector function. This, together with all the scalar multiples of $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ give all the equilibrium points of the SDE.

4.4.11 Stability of Equilibrium Points

Another way of classifying equilibrium points is in terms of their stability. Stability concepts are fundamental in engineering and other applications.

An equilibrium point O of a system is called stable if all trajectories of the system that are close to O at some instant t_0 remain close to O at all future times $t > t_0$. Otherwise O is called unstable.

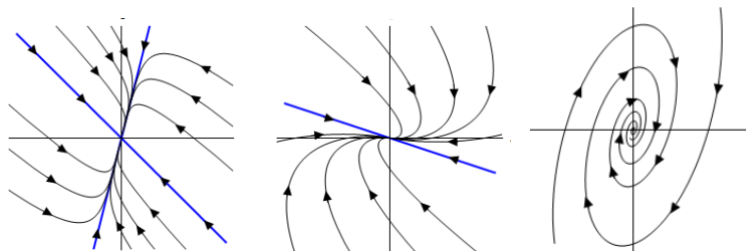
Stability ➔

a small change/disturbance of a physical system at some instant changes the behavior of the system only slightly at all future times t

There are three types of stability, namely asymptotically stable, neutrally stable, and unstable, and they can also be determined by the eigenvalues of the underlying matrix of the SDE.

4.4.12 Asymptotically stable

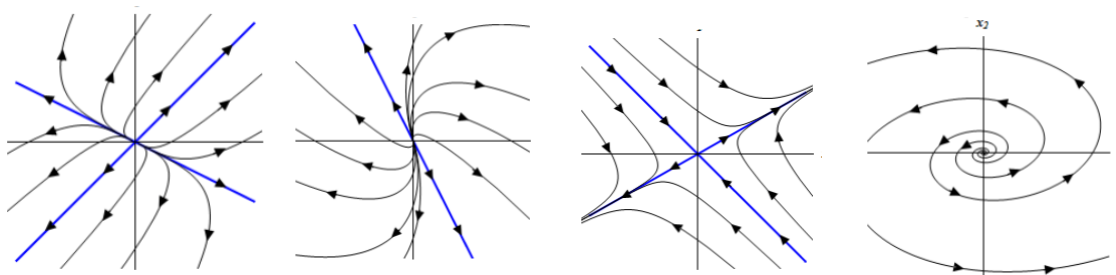
An equilibrium point O is called asymptotically stable if every trajectory that has a point near O gets closer and closer to O as t increases.



The diagrams above show nodes, degenerate nodes and spiral which are asymptotically stable. A star node is asymptotically stable if all its trajectories are pointing towards the equilibrium point.

4.4.13 Unstable

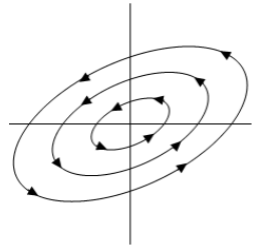
An equilibrium point O is called unstable if at least one trajectory that has a point near O gets further away from O as t increases.



The diagrams above show nodes, degenerate node, saddle point and spiral which are unstable. Similarly, a star node is unstable if all its trajectories are pointing away from the equilibrium point.

4.4.14 Neutrally stable

The only type of equilibrium point that is neutrally stable is the center. The trajectories will just move around the equilibrium point and never actually move towards it as t increases.



4.4.15 Eigenvalue Criteria

Let r_1, r_2 be the two eigenvalues of the underlying 2×2 matrix $\mathbf{A}$ of the SDE.

Types of equilibrium points	Stability	Eigenvalue Conditions
Node, Degenerate node, Star node	Asymptotically stable	r_1 and r_2 both negative
	Unstable	r_1 and r_2 both positive
Saddle point	(Always) unstable	r_1 and r_2 opposite signs
Center	(Always) Neutrally stable	r_1 and r_2 purely imaginary
Spiral point	Asymptotically stable	Real parts of r_1 and r_2 both negative
	Unstable	Real parts of r_1 and r_2 both positive

4.4.16 Mass - Spring System

Recall that the mass-spring system is given by this 2nd order ODE: $my'' + cy' + ky = 0$

which can be converted into the SDE: $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

Here k : Spring constant and c : damping constant. We may assume k and $c > 0$ except when there is no damping ($c = 0$).

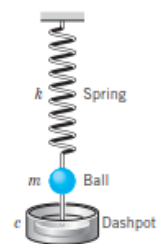
We would like to find out the type of equilibrium point and its stability, under various conditions. In this context, the equilibrium point represents the stage when the ball is at rest.

The characteristic polynomial of the matrix $\begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}$ is given by:

$$\begin{vmatrix} \lambda & -1 \\ \frac{k}{m} & \lambda + \frac{c}{m} \end{vmatrix} = \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m}$$

By solving for the roots of this quadratic polynomial, we get the eigenvalues r_1 and r_2 :

$$\frac{1}{2} \left(-\frac{c}{m} \pm \sqrt{\left(\frac{c}{m}\right)^2 - \frac{4k}{m}} \right).$$



Let's simplify the expression by denoting

$$\Delta = \left(\frac{c}{m}\right)^2 - \frac{4k}{m}$$

So we have $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$.

We will consider a few cases:

Case 1: $c = 0$ (no damping)

In this case, $\frac{c}{m} = 0$ and $\Delta < 0$. So $r_1 = \frac{1}{2}\sqrt{\Delta}$ and $r_2 = -\frac{1}{2}\sqrt{\Delta}$ are purely imaginary complex eigenvalues. The equilibrium point is a center, which is neutrally stable.

Case 2: $c^2 > 4mk$

In this case, $\Delta > 0$. So $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$ are real distinct eigenvalues. Furthermore, both are negative numbers (see below *). The equilibrium point is a node and is asymptotically stable.

(*Clearly r_2 is negative. To see r_1 is also negative, note that $\Delta = \left(\frac{c}{m}\right)^2 - \frac{4k}{m} < \left(\frac{c}{m}\right)^2 \Rightarrow \sqrt{\Delta} < \frac{c}{m}$ by taking square root on both sides. Hence $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right) < 0$.)

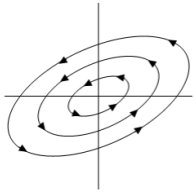
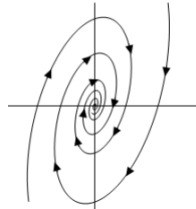
Case 3: $c^2 = 4mk$

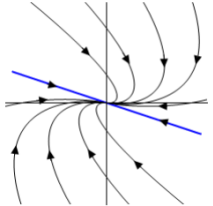
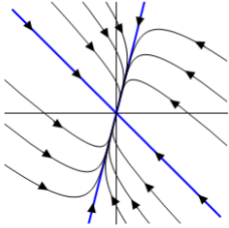
In this case, $\Delta = 0$. So $r_1 = r_2 = -\frac{c}{2m}$ is a real repeated negative eigenvalue. The equilibrium point is a degenerate node and is asymptotically stable.

Case 4: $0 < c^2 < 4k$

In this case, $\frac{c}{m} > 0$ and $\Delta < 0$. So $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$ are complex eigenvalues with negative real parts. The equilibrium point is a spiral and is asymptotically stable.

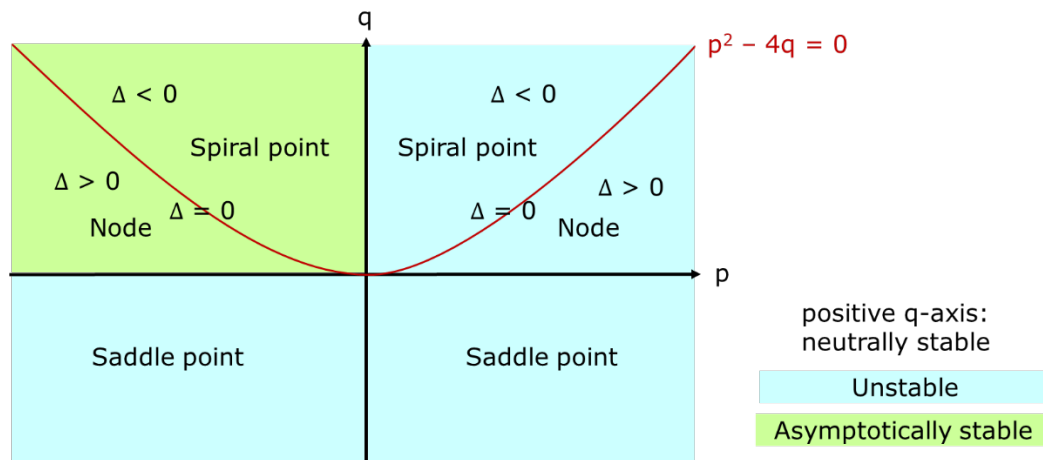
We summarize the 4 cases as follow:

Damping	Condition	Type of points	Trajectories
No damping (the ball will keep oscillating without coming to a halt)	$c = 0$	Centre	
Underdamping (the damping force is weak relative to the spring force; the ball will oscillate with smaller and smaller amplitude before coming to a stop.)	$c^2 < 4mk$	Spiral point	

Critical damping (the ball will come to rest as quickly as possible without oscillation)	$c^2 = 4mk$	Degenerate node	
Over damping (the damping force is strong relative to the spring force; the ball again will return to the equilibrium state without oscillating; the stronger the damping force, the slower will the ball come to rest)	$c^2 > 4mk$	Node	

4.4.17 Stability Chart (Optional)

To end this section, here is the stability chart that is commonly used for the classification of the type and stability of equilibrium points.



This is the p - q plane with p the horizontal axis and q the vertical axis ($p = r_1 + r_2$, $q = r_1 r_2$). The two axes divide the plane into 4 quadrants. The first quadrant is when p and q are both positive; the second quadrant is for negative p and positive q ; and the third and fourth quadrants are for negative q .

The colours indicate the stability. So the first, third and fourth quadrants are the unstable zone, while the second quadrant is the asymptotically stable zone. The positive q -axis between the first and second quadrants is the neutrally stable zone.

To indicate the type of equilibrium points, we include a quadratic curve given by the equation $p^2 - 4q = 0$. Here we denote $\Delta = p^2 - 4q$. This is the line with the condition $\Delta = 0$, and is where all the degenerate nodes lie.

This curve further divides each of the first two quadrants into two parts. Above the curve, $\Delta < 0$ and hence all the spiral points are found here, except the q -axis, which consists of the center. Below the curve, $\Delta > 0$. All the nodes can be found between the curve and the p -axis.

Finally, everything below the p -axis are saddle points.

The stability chart provides us with some insight how the phase portrait transforms when we move from one zone to another zone on the stability chart.

4.5. Real Distinct Eigenvalues

In the last three sections, we shall look more closely into the solutions of 2 x 2 SDE based on the eigenvalues of the underlying matrices. We start with matrices with two real eigenvalues that are distinct.

4.5.1 Eigenvalue Criteria

Eigenvalues r_1 and r_2 (distinct)	Equilibrium Points	Stability
r_1 and $r_2 > 0$	Node	Unstable
r_1 and $r_2 < 0$	Node	Asymptotically stable
$r_1 > 0$ and $r_2 < 0$	Saddle point	Unstable

4.5.2 Real and Opposite Sign

$$\begin{cases} y_1' &= y_1 + 2y_2 \\ y_2' &= 3y_1 + 2y_2 \end{cases}$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$:

The characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda-1 & -2 \\ -3 & \lambda-2 \end{vmatrix} = \lambda^2 - 3\lambda - 4$.

We get two distinct eigenvalues $r_1 = 4$ and $r_2 = -1$ of opposite sign

and eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ for r_1 and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2 . [Again here we are taking a scalar multiple for the eigenvector $\mathbf{v}_1$ to avoid fraction in the entries.]

So the general solution:

$$\mathbf{y} = c_1 e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 2e^{4t} \\ 3e^{4t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-t} \\ e^{-t} \end{pmatrix}$$

Alternatively, we can present the solutions in terms of the two component functions:

$$y_1 = 2c_1 e^{4t} - c_2 e^{-t}, \quad y_2 = 3c_1 e^{4t} + c_2 e^{-t}.$$

Trajectories

To sketch the solutions on the phase plane, the two eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ will be needed as a guide.

By specifying the values for the parameters c_1 and c_2 , we will get a specific trajectory.

Case $c_2 = 0$:

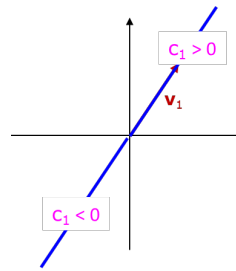
The solutions become

$$y_1 = 2c_1 e^{4t}, \quad y_2 = 3c_1 e^{4t}$$

By eliminating the common exponential functions from the parametric equations, we have $y_2 = 3/2 y_1$ which is a straight line parallel to the eigenvector $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ (see diagram below).

If $c_1 > 0$, the trajectory is in 1st quadrant since both y_1 and y_2 are positive in this case.

If $c_1 < 0$, the trajectory is in 3rd quadrant since both y_1 and y_2 are negative in this case.



Case $c_1 = 0$:

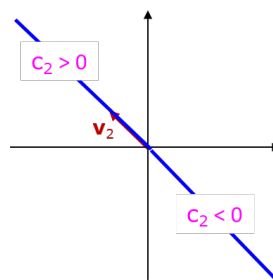
The solutions become

$$y_1 = -c_2 e^{-t}, \quad y_2 = c_2 e^{-t}$$

which gives us this equation $y_2 = -y_1$, a straight line parallel to the eigenvector $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

If $c_2 > 0$, the trajectory is in 2nd quadrant since y_1 is negative and y_2 is positive.

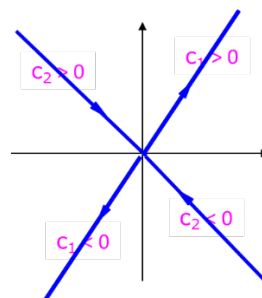
If $c_2 < 0$, the trajectory is in 4th quadrant since y_1 is positive and y_2 is negative.



Now we look at the direction of the trajectories.

When $c_1 = 0$, we observe that, as t increases, the exponential function (e^{-t}) will get smaller and smaller, and hence the trajectory will be moving towards the origin.

On the other hand, for $c_2 = 0$, as t increases, the exponential function (e^{4t}) will get larger and larger, and hence the trajectory will be moving away from the origin.



Case both $c_1, c_2 \neq 0$: We will have a combination of these behaviors.

Large negative t 's represent the beginning stage of the trajectories:

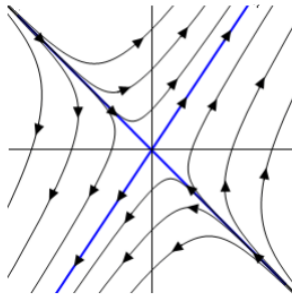
For such values of t , the solution will be dominated by the e^{-t} term since the exponent for this term will be large and positive with negative values of t . Trajectories for large negative t will be parallel to $\mathbf{v}_2$ and moving in the direction towards the origin.

Large positive t 's represent the ending stage of the trajectories:

In this case, the solution will be dominated by the e^{4t} term since the exponent for this term will be large and positive with positive values of t . Trajectories in this case will be parallel to $\mathbf{v}_1$ and moving away from the origin.

Putting the above together, a general trajectory will start moving parallel to $\mathbf{v}_2$ towards the origin, and as they get closer to the origin they will gradually turn away towards $\mathbf{v}_1$ and then continue up parallel to this vector.

Sketching some of these trajectories in the phase plane will give us the phase portrait.



In this case, the equilibrium point is a saddle point and is unstable.

Initial Value Problem

Let's solve an initial value problem with this SDE.

$$\begin{cases} y_1' = y_1 + 2y_2 \\ y_2' = 3y_1 + 2y_2 \end{cases} \quad \text{with initial condition: } \begin{cases} y_1(0) = 0 \\ y_2(0) = -4 \end{cases}$$

In matrix form:

$$\mathbf{y}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{y} \quad \mathbf{y}(0) = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$$

The general solution is given by

$$\mathbf{y} = c_1 \begin{pmatrix} 2e^{4t} \\ 3e^{4t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-t} \\ e^{-t} \end{pmatrix}$$

By substituting $t = 0$ into the general solution above, we have

$$\mathbf{y}(0) = \begin{pmatrix} 0 \\ -4 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

This becomes a regular linear system which we can easily solve $c_1 = -\frac{4}{5}$, $c_2 = -\frac{8}{5}$.

This gives us the particular solution

$$\mathbf{y} = -\frac{4}{5} \begin{pmatrix} 2e^{4t} \\ 3e^{4t} \end{pmatrix} - \frac{8}{5} \begin{pmatrix} -e^{-t} \\ e^{-t} \end{pmatrix}$$

In term of the component functions, we have

$$y_1(t) = -\frac{8}{5}e^{4t} + \frac{8}{5}e^{-t} \quad y_2(t) = -\frac{12}{5}e^{4t} + \frac{8}{5}e^{-t}.$$

4.5.3 Real and Same Sign

$$\begin{cases} y_1' &= -5y_1 + y_2 \\ y_2' &= 4y_1 - 2y_2 \end{cases}$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix}$.

The characteristic polynomial is $\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda+5 & -1 \\ -4 & \lambda+2 \end{vmatrix} = \lambda^2 + 7\lambda + 6$.

So we have eigenvalues $r_1 = -1$ and $r_2 = -6$ which are both negative,

and the corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2 .

[For the same reason as above, we use a scalar multiple for the eigenvector $\mathbf{v}_1$.]

The general solution is given by

$$\mathbf{y} = c_1 \begin{pmatrix} e^{-t} \\ 4e^{-t} \end{pmatrix} + c_2 \begin{pmatrix} -e^{-6t} \\ e^{-6t} \end{pmatrix}.$$

In terms of the components y_1 and y_2 of the solution, we have

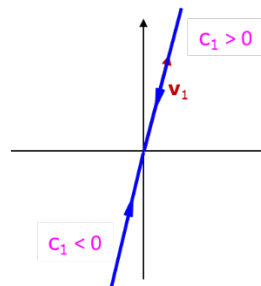
$$y_1 = c_1 e^{-t} - c_2 e^{-6t} \quad \text{and} \quad y_2 = 4c_1 e^{-t} + c_2 e^{-6t}.$$

Trajectories

Case $c_2 = 0$: trajectory is a straight line parallel to $\mathbf{v}_1$

If $c_1 > 0$, the trajectory is in 1st quadrant; if $c_1 < 0$, the trajectory is in 3rd quadrant.

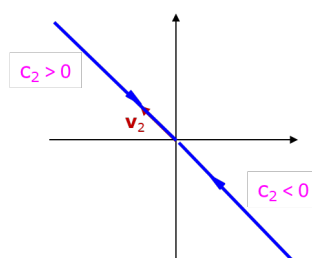
Since the eigenvalue is negative, as t increases the exponential function (e^{-t}) will get smaller and smaller and hence the trajectory is moving towards the origin.



Case $c_1 = 0$: trajectory is a straight line parallel to $\mathbf{v}_2$

If $c_2 > 0$, the trajectory is in 2nd quadrant; if $c_2 < 0$, the trajectory is in 4th quadrant.

As t increases the exponential function (e^{-6t}) will decrease and the trajectory will approach the origin.



Case both $c_1, c_2 \neq 0$: the behavior is quite different from the previous example.

Note that both our eigenvalues are negative with $r_1 > r_2$.

For large negative t :

e^{-t} will be smaller than e^{-6t} . Hence the solution will be dominated by the e^{-6t} term (since the exponent for this term will have a larger positive value).

This means the trajectories will be parallel to $\mathbf{v}_2$ at the beginning, and move in the direction towards the origin.

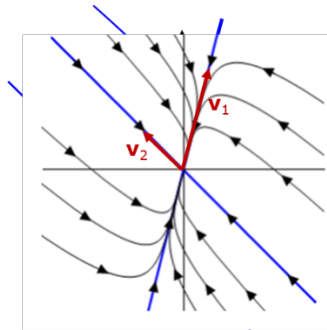
For large positive t :

e^{-6t} will be smaller than e^{-t} . The solution will be dominated by the e^{-t} term (since the exponent for this term will have a larger positive value).

The trajectories in this case will be parallel to $\mathbf{v}_1$ as it moves towards the origin.

Putting together, a general trajectory will start away and parallel to $\mathbf{v}_2$, move in towards the origin, and they will gradually become parallel to $\mathbf{v}_1$ as they converge to the origin.

Sketching some of these trajectories in the phase plane will give us the phase portrait



In this case, the equilibrium point is a node and is asymptotically stable.

4.6. Complex Eigenvalues

In this section, we look at the solution of 2 x 2 SDE with underlying matrix having complex eigenvalues. Pay attention how to construct real value solutions for such system.

4.6.1 Eigenvalue Criteria

Eigenvalues $a + bi, a - bi$	Equilibrium Points	Stability
$a > 0$	Spiral	Unstable
$a < 0$	Spiral	Asymptotically stable
$a = 0$ (purely imaginary)	Center	Neutrally stable

Complex eigenvalues always come in conjugate pairs: $a + bi$ and $a - bi$

The corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} c + d i \\ e + f i \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c - d i \\ e - f i \end{pmatrix}$ are also complex conjugate of each other.

We can also write the eigenvectors as $\mathbf{v}_1 = \mathbf{p} + \mathbf{q} i$ and $\mathbf{v}_2 = \mathbf{p} - \mathbf{q} i$ with real part vector $\mathbf{p} = \begin{pmatrix} c \\ e \end{pmatrix}$ and imaginary part vector $\mathbf{q} = \begin{pmatrix} d \\ f \end{pmatrix}$.

4.6.2 Complex Eigenvalues

$$\begin{cases} y_1' &= 3y_1 - 13y_2 \\ y_2' &= 5y_1 + y_2 \end{cases}$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} 3 & -13 \\ 5 & 1 \end{pmatrix}$ has characteristic polynomial

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda - 3 & 13 \\ -5 & \lambda - 1 \end{vmatrix} = \lambda^2 - 4\lambda + 68$$

which gives a pair of eigenvalues which are complex conjugate: $2+8i$ and $2-8i$.

To find eigenvector $\mathbf{v}_1$ corresponding to $r_1 = 2 + 8i$, we will solve the homogeneous system

$$((2+8i)\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$$

In standard form:

$$\begin{cases} (-1 + 8i)x + 13y = 0 & (1) \\ -5x + (1 + 8i)y = 0 & (2) \end{cases}$$

When a 2 x 2 linear system arises from solving for eigenspace, the two equations in the system will always be multiple of each, and we only need to work on one of them.

In our case, check that: $(1 + 8i) \times (1) = 13 \times (2)$.

This means we just have to solve any one of the two equations.

Let's work on equation (2) and set $y = t$ (parameter). Then we have $x = \frac{1}{5}(1+8i)t$.

So we have the general solution: $\begin{cases} x &= \frac{1}{5}(1+8i)t \\ y &= t \end{cases}$

By letting $t = 1$, we obtain a particular eigenvector $\mathbf{v} = \begin{pmatrix} \frac{1}{5} + \frac{8}{5}i \\ 1 \end{pmatrix}$.

In fact, to avoid fractions in the components, we can substitute $t = 5$ in the general solution to get the eigenvector $\mathbf{v}_1 = \begin{pmatrix} 1 + 8i \\ 5 \end{pmatrix}$.

For the other eigenvalue $2-8i$, we do not need to go through the same procedure again to find its eigenvector, as this eigenvector $\mathbf{v}_2$ will simply be the complex conjugate of $\mathbf{v}_1$, i.e. $\mathbf{v}_2 = \begin{pmatrix} 1 - 8i \\ 5 \end{pmatrix}$.

Now we have found all the eigenvalues and eigenvectors of the matrix. However, we cannot use these complex values to construct real solutions of the system. We shall have a general discussion in the next segment on how to convert complex solutions to real solutions of the system, before coming back to solve this SDE.

4.6.3 Complex Solutions

Recall that solutions of the SDE are given by

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{r_1 t}, \quad \mathbf{x}_2(t) = \mathbf{v}_2 e^{r_2 t}$$

But if our eigenvalues $r_1 = a + bi$ and $r_2 = a - bi$ and eigenvectors $\mathbf{v}_1 = \mathbf{p} + \mathbf{q}i$, and $\mathbf{v}_2 = \mathbf{p} - \mathbf{q}i$ are complex, this will give us complex solutions of the system, which is not what we want.

We need to use Euler's formula to obtain real solutions from this complex solutions:

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

So $e^{a+bi} = e^a e^{bi} = e^a (\cos b + i \sin b)$

$$\begin{aligned} \text{Then } \mathbf{x}_1(t) &= \mathbf{v}_1 e^{r_1 t} = (\mathbf{p} + \mathbf{q}i) e^{(a+bi)t} \\ &= (\mathbf{p} + \mathbf{q}i) e^{at} (\cos bt + i \sin bt) \\ &= e^{at} (\mathbf{p} \cos bt - \mathbf{q} \sin bt) + e^{at} (\mathbf{p} \sin bt + \mathbf{q} \cos bt) i \quad (\#) \end{aligned}$$

Taking the real and imaginary parts of (#) separately:

$$\begin{aligned} \mathbf{x}_{\text{Re}}(t) &= e^{at} (\mathbf{p} \cos bt - \mathbf{q} \sin bt) \\ \mathbf{x}_{\text{Im}}(t) &= e^{at} (\mathbf{p} \sin bt + \mathbf{q} \cos bt). \end{aligned}$$

We shall see that $\mathbf{x}_{\text{Re}}(t)$ and $\mathbf{x}_{\text{Im}}(t)$ are real solutions for the system.

First, observe the following:

Starting with the eigenvector $\mathbf{v}_1 = \mathbf{p} + \mathbf{q}i$, if we pre-multiply it with $\mathbf{A}$, we will get

$$\mathbf{A}\mathbf{v}_1 = \mathbf{A}\mathbf{p} + \mathbf{A}\mathbf{q}i \quad (*)$$

where $\mathbf{A}\mathbf{p}$ and $\mathbf{A}\mathbf{q}$ are the real part and imaginary part.

On the other hand, since $\mathbf{v}_1$ is an eigenvector of $\mathbf{A}$, so

$$\mathbf{A}\mathbf{v}_1 = r_1 \mathbf{v}_1 = (a + bi)(\mathbf{p} + \mathbf{q}i) = (\mathbf{a}\mathbf{p} - \mathbf{b}\mathbf{q}) + (\mathbf{b}\mathbf{p} + \mathbf{a}\mathbf{q})i \quad (**)$$

Comparing the real parts for both expressions (*) and (**) of $\mathbf{A}\mathbf{v}_1$, we get

$$\mathbf{A}\mathbf{p} = \mathbf{a}\mathbf{p} - \mathbf{b}\mathbf{q}. \quad (\text{I})$$

Similarly, comparing the imaginary parts for both expressions, we get

$$\mathbf{A}\mathbf{q} = \mathbf{b}\mathbf{p} + \mathbf{a}\mathbf{q}. \quad (\text{II})$$

4.6.4 Real Solutions

We now show

$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt)$$

are solutions of the SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

Differentiating $\mathbf{x}_{\text{Re}}(t)$ using product rule, we get

$$\mathbf{x}_{\text{Re}}'(t) = ae^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt) + be^{at}(-\mathbf{p} \sin bt - \mathbf{q} \cos bt) \quad (\#)$$

On the other hand, pre-multiplying $\mathbf{x}_{\text{Re}}(t)$ by $\mathbf{A}$ gives us

$$\mathbf{A}\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{A}\mathbf{p} \cos bt - \mathbf{A}\mathbf{q} \sin bt)$$

By replacing $\mathbf{A}\mathbf{p}$ and $\mathbf{A}\mathbf{q}$ in the right hand side using (I) and (II) in the previous segment, we get

$$\mathbf{A}\mathbf{x}_{\text{Re}}(t) = e^{at}((a\mathbf{p} - b\mathbf{q})\cos bt - (b\mathbf{p} + a\mathbf{q})\sin bt) \quad (\Delta)$$

We compare the right hand sides of (#) and (Δ), and check that they are the same. Hence we conclude that

$$\mathbf{x}_{\text{Re}}'(t) = \mathbf{A}\mathbf{x}_{\text{Re}}(t).$$

This means $\mathbf{x}_{\text{Re}}(t)$ is indeed a solution of our system $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

Likewise, $\mathbf{x}_{\text{Im}}(t)$ can be shown in a similar way to be a solution of the system.

Furthermore, the two vector functions $\mathbf{x}_{\text{Re}}(t)$ and $\mathbf{x}_{\text{Im}}(t)$ are not scalar multiple of each other. Therefore, the two solutions are linearly independent and hence form a fundamental set of solutions for the SDE, and we can write the general solution of the system as

$$\mathbf{y} = c_1\mathbf{x}_{\text{Re}}(t) + c_2\mathbf{x}_{\text{Im}}(t).$$

4.6.5 General Solutions with Complex Eigenvalues

Step 1. For one eigenvalue $r_1 = a + bi$, find its corresponding eigenvector $\mathbf{v}_1$.

The second eigenvalue r_2 and its eigenvector $\mathbf{v}_2$ are complex conjugates of the first.

The eigenvectors are of the form $\mathbf{v}_1 = \mathbf{p} + \mathbf{q}i$, $\mathbf{v}_2 = \mathbf{p} - \mathbf{q}i$.

Step 2. Construct the linearly independent real ($\mathbf{x}_{\text{Re}}$) and imaginary ($\mathbf{x}_{\text{Im}}$) parts of the solutions as follows:

$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt)$$

This gives a fundamental set of solutions to the SDE.

Step 3. The general (real) solution is $\mathbf{y} = c_1\mathbf{x}_{\text{Re}}(t) + c_2\mathbf{x}_{\text{Im}}(t)$

4.6.6 Example 1: Complex Conjugates

Let's go back to our example with matrix $\mathbf{A} = \begin{pmatrix} 3 & -13 \\ 5 & 1 \end{pmatrix}$.

We shall write out $\mathbf{x}_{\text{Re}}(t)$ and $\mathbf{x}_{\text{Im}}(t)$ explicitly.

Using the eigenvalue $r_1 = 2 + 8i$, we have $a = 2$ and $b = 8$.

The corresponding eigenvector $\mathbf{v}_1 = \begin{pmatrix} 1 + 8i \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} + \begin{pmatrix} 8 \\ 0 \end{pmatrix} i$, and so we have

$$\mathbf{p} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \text{ and } \mathbf{q} = \begin{pmatrix} 8 \\ 0 \end{pmatrix}.$$

Therefore, our two basis solutions:

$$\mathbf{x}_{\text{Re}}(t) = e^{2t} \left(\begin{pmatrix} 1 \\ 5 \end{pmatrix} \cos 8t - \begin{pmatrix} 8 \\ 0 \end{pmatrix} \sin 8t \right)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{2t} \left(\begin{pmatrix} 1 \\ 5 \end{pmatrix} \sin 8t + \begin{pmatrix} 8 \\ 0 \end{pmatrix} \cos 8t \right)$$

The general solution of the system is given by

$$\mathbf{y} = c_1 e^{2t} \begin{pmatrix} \cos(8t) - 8\sin(8t) \\ 5\cos(8t) \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} \sin(8t) + 8\cos(8t) \\ 5\sin(8t) \end{pmatrix}.$$

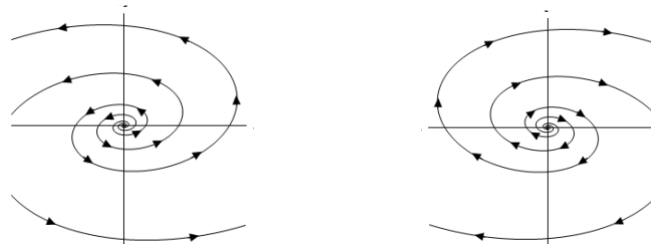
Since the exponential function is a common term for both basis solutions, we can also write the general solution as

$$\mathbf{y} = e^{2t} \left[c_1 \begin{pmatrix} \cos(8t) - 8\sin(8t) \\ 5\cos(8t) \end{pmatrix} + c_2 \begin{pmatrix} \sin(8t) + 8\cos(8t) \\ 5\sin(8t) \end{pmatrix} \right].$$

We shall now sketch a trajectory of the system.

In the general solution above, the terms inside the square brackets involves only sine and cosine functions, which are periodic. The effect is a curve that will revolve about the origin. The part that control the expansion of the trajectory is the exponential term e^{2t} .

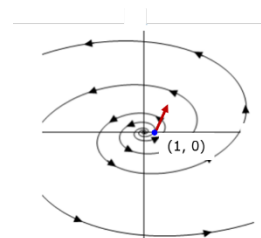
Since the real part of the eigenvalue $r_1 = 2 + 8i$ is positive ($a = 2$), the trajectories will spiral out from the origin. And we know the equilibrium point is a spiral which is unstable.



The only thing left to check is the direction of rotation. How do we tell whether the trajectory looks like the one on the left or the right? In other words, whether the direction is clockwise or counterclockwise.

To determine that, we just need to test a point on the positive horizontal axis, say $(1, 0)$:

- Pre-multiply $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ by matrix $\mathbf{A} = \begin{pmatrix} 3 & -13 \\ 5 & 1 \end{pmatrix}$ to get $\begin{pmatrix} 3 & -13 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$;
- Then $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ is the tangent vector to the trajectory at point $(1, 0)$;
- Since $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ is pointing upwards at the point $(1, 0)$ and the trajectory spirals about the origin, it means the direction is counterclockwise.



4.6.7 Example 2: Purely Imaginary

$$\begin{cases} y_1' &= 3y_1 - 9y_2 \\ y_2' &= 4y_1 - 3y_2 \end{cases}$$

The underlying matrix is $\mathbf{A} = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix}$.

The characteristic polynomial $\det(\lambda \mathbf{I} - \mathbf{A}) = \begin{vmatrix} \lambda-3 & 9 \\ -4 & \lambda+3 \end{vmatrix} = \lambda^2 + 27$ gives purely imaginary eigenvalues

$$r_1 = 3\sqrt{3}i, \quad r_2 = -3\sqrt{3}i$$

So $a = 0$, $b = 3\sqrt{3}$

To find eigenvector $\mathbf{v}_1$, we solve the system $(3\sqrt{3}i \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$. This gives the two linear equations

$$\begin{cases} (3\sqrt{3}i - 3)x + 9y = 0 & (1) \\ -4x + (3\sqrt{3}i + 3)y = 0 & (2) \end{cases}$$

As before, the two equations (1) and (2) are multiples of each other: $(3\sqrt{3}i + 3) \times (1) = 9 \times (2)$.

So we only need to work on one of them, say (1).

Set $y = t$ (parameter). Then $x = \frac{3}{1-\sqrt{3}i} t$. (We can also set x as parameter and solve for y . See ** below)

$$\text{General Solution: } \begin{cases} x &= \frac{3}{1-\sqrt{3}i} t \\ y &= t \end{cases}$$

Let's choose an appropriate value $t = 1-\sqrt{3}i$ so that the eigenvector $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 1-\sqrt{3}i \end{pmatrix}$ is free of fraction.

****** If we set in equation (1) $x = t$ (parameter), then $y = \frac{1-\sqrt{3}i}{3} t$ and General Solution: $\begin{cases} x &= t \\ y &= \frac{1-\sqrt{3}i}{3} t \end{cases}$.

By choosing $t = 3$, we get the same eigenvector $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 1-\sqrt{3}i \end{pmatrix}$ above.

The eigenvector for the other eigenvalue $r_2 = -3\sqrt{3}i$ is given by the complex conjugate $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 1+\sqrt{3}i \end{pmatrix}$.

From the eigenvector $\mathbf{v}_1$, we have $\mathbf{p} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 0 \\ -\sqrt{3} \end{pmatrix}$.

Substituting all the values into the two real solutions

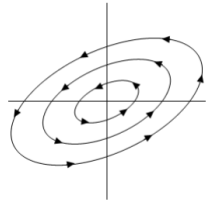
$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt)$$

we get the general solution of the system:

$$\mathbf{y} = c_1 \begin{pmatrix} 3\cos(3\sqrt{3}t) \\ \cos(3\sqrt{3}t) + \sqrt{3}\sin(3\sqrt{3}t) \end{pmatrix} + c_2 \begin{pmatrix} 3\sin(3\sqrt{3}t) \\ \sin(3\sqrt{3}t) - \sqrt{3}\cos(3\sqrt{3}t) \end{pmatrix}$$

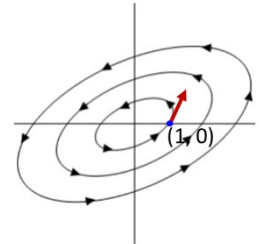
Since the eigenvalues are purely imaginary, the exponential term is absent and the trajectories of the solutions will be circles or ellipses that are centered at the origin. And the equilibrium point is a center that is always neutrally stable.



The only thing left to check is whether the trajectories are rotating in a clockwise or counterclockwise direction.

As before, we can test this by taking a point on the positive horizontal axis, say $(1, 0)$.

- Pre-multiply $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ by $\mathbf{A} = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix}$ to get $\begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.
- $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is the tangent vector to the trajectory at the point $(1, 0)$
- Since $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is pointing upwards at $(1, 0)$, the trajectories are moving in counterclockwise direction.



4.7. Repeated Eigenvalues

This last case is when the underlying matrix of a system has a repeated eigenvalue. For such cases, it is likely that we can only find one linearly independent eigenvector. This in turn means that we will not be able to construct two linearly independent solutions for the system in the usual way. In order to find a fundamental set of solutions for the SDE, we have to introduce the notion of generalized eigenvector.

4.7.1 Eigenvalue-Eigenvector Criteria

Eigenvalue r (double root)	Eigenvectors	Equilibrium Points	Stability
$r > 0$	Two independent eigenvectors	Star node	Unstable
$r < 0$			Asymptotically stable
$r > 0$	Only one independent eigenvector	Degenerate node	Unstable
$r < 0$			Asymptotically stable

When the characteristic polynomial of the matrix $\mathbf{A}$ has a double root, we have a repeat eigenvalue r . For such matrices, there are two possibilities:

- (i) There are two linearly independent eigenvectors corresponding to the eigenvalue r . In this case, $\mathbf{A}$ is diagonalizable. The equilibrium point is a star node. It can be stable or unstable depending on the sign of the eigenvalue.
- (ii) There is only one linearly independent eigenvector corresponding to r . In this case, $\mathbf{A}$ is not diagonalizable. The equilibrium point is a degenerate node. Again the stability depends on the sign of the eigenvalue.

4.7.2 Example 1: Two Linearly Independent Eigenvectors

$$\begin{cases} y_1' &= 3y_1 \\ y_2' &= 3y_2 \end{cases}$$

This is a decoupled system as each equation only involved one function y_1 or y_2 . In other words, the two functions are not dependent of each other.

In this case, the matrix $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ is a diagonal matrix. This is called a scalar matrix as both diagonal entries are the same. Since this is a diagonal matrix, we can read off the eigenvalue from the diagonal entries, which is 3, and it is a repeated eigenvalue.

Note that for 2×2 scalar matrices, every vector in $\mathbf{R}^2$ is an eigenvector. In particular, if we take $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, we get two linearly independent eigenvectors for $\mathbf{A}$, both associated to the eigenvalue 3.

The general solution:

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{3t} = c_1 \begin{pmatrix} e^{3t} \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^{3t} \end{pmatrix} = \begin{pmatrix} c_1 e^{3t} \\ c_2 e^{3t} \end{pmatrix}$$

Note that in this case, only one exponential function e^{3t} is involved in the solution.

Trajectories

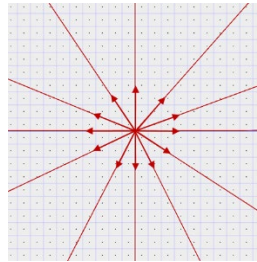
The two component solutions y_1 and y_2 of the above system are given by

$$y_1 = c_1 e^{3t} \quad \text{and} \quad y_2 = c_2 e^{3t}$$

Since they are both scalars of the same exponential function e^{3t} , we can easily eliminate t from the two equations to get

$$y_2 = \frac{c_2}{c_1} y_1$$

which is the equation of a line on the phase plane. Hence the trajectories are all straight lines through the origin. Since c_1 and c_2 can be any value, the gradient $\frac{c_2}{c_1}$ of the trajectories can be any value as well. Hence the phase portrait of the system looks like this:



As the eigenvalue ($r = 3$) is positive, all the trajectories will start near the origin and move away from it as t increases. Hence the equilibrium point is a star node which is unstable.

4.7.3 Example 2: Only One Independent Eigenvector

$$\begin{cases} y_1' &= 2y_1 - y_2 \\ y_2' &= 4y_1 + 6y_2 \end{cases}$$

The underlying matrix $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & 6 \end{pmatrix}$ is not diagonalizable:

We check that it has only one eigenvalue 4 with only one linearly independent eigenvector $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

(All other eigenvectors of $\mathbf{A}$ are just scalar multiple of $\mathbf{v}$, and hence we can only find one linearly independent eigenvector.)

So using our usual approach of constructing solution for the SDE, we can only find one linearly independent solution:

$$\mathbf{x}_1 = e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

However, for a 2×2 system we need another solution to form a fundamental set of solutions.

A second (linearly independent) solution to the system must involve another vector, called a generalized eigenvector of $\mathbf{A}$. This is not really an eigenvector, but it can be used to construct an independent solution of the SDE.

4.7.3a Finding a Second Solution

The trick is to consider a second solution $\mathbf{x}_2$ given in the following form:

$$\mathbf{x}_2(t) = te^{4t} \mathbf{v} + e^{4t} \mathbf{u} \quad (\boxtimes)$$

Here $\mathbf{u}$ is a vector yet to be determined, while $\mathbf{v}$ is our sole independent eigenvector. Notice that the two terms in $\mathbf{x}_2$ are similar to our usual solution, except there is an extra t in the first term $te^{4t} \mathbf{v}$.

We need to find a suitable vector $\mathbf{u}$ in order for $\mathbf{x}_2$ to be a solution. This vector $\mathbf{u}$ can be obtained by solving the non-homogeneous system

$$(\mathbf{A} - 4\mathbf{I})\mathbf{u} = \mathbf{v} \quad (\Delta)$$

Note that $\mathbf{u}$ is dependent on the eigenvector $\mathbf{v}$ and its eigenvalue 4.

To check that $\mathbf{x}_2$ in $(\boxtimes)$ is a solution of our SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$:

LHS: Differentiate $\mathbf{x}_2$ in $(\boxtimes)$:

$$\mathbf{x}_2'(t) = e^{4t} \mathbf{v} + 4te^{4t} \mathbf{v} + 4e^{4t} \mathbf{u} \quad (1)$$

Note that we have applied product rule on the right hand side to the term $te^{4t} \mathbf{v}$. The two vectors $\mathbf{u}$ and $\mathbf{v}$ are independent of t , and are treated as constants under differentiation.

RHS: Multiply $\mathbf{A}$ to $\mathbf{x}_2$:

$$\mathbf{A}\mathbf{x}_2 = \mathbf{A}(te^{4t} \mathbf{v} + e^{4t} \mathbf{u}) = te^{4t} \mathbf{A}\mathbf{v} + e^{4t} \mathbf{A}\mathbf{u} = 4te^{4t} \mathbf{v} + e^{4t} (4\mathbf{u} + \mathbf{v})$$

which is the same (1). The last equation above is derived from (Δ) by substituting $\mathbf{A}\mathbf{u} = 4\mathbf{u} + \mathbf{v}$.

Hence we have shown $\mathbf{x}_2$ is a solution of the SDE. Note that this solution is independent of $\mathbf{x}_1$.

4.7.3b Generalized Eigenvector

We proceed to solve this matrix equation $(\mathbf{A} - 4\mathbf{I})\mathbf{u} = \mathbf{v}$ (Δ) in our earlier example:

$$\begin{pmatrix} -2 & -1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

[Note: We cannot replace (Δ) by solving for $\mathbf{u}$ using $(4\mathbf{I} - \mathbf{A})\mathbf{u} = \mathbf{v}$. This will lead to wrong solution for the SDE.]

Using Gaussian elimination or otherwise, we can get a general solution

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(1-k) \\ k \end{pmatrix} = k \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

where k is a parameter.

In other words, there are many choices for our vector $\mathbf{u}$. These vectors are known as generalized eigenvectors of $\mathbf{A}$. We just need one of them. Choose $k = 0$ for a specific solution $\mathbf{u} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$.

We are ready to substitute this $\mathbf{u}$ into

$$\mathbf{x}_2(t) = te^{4t} \mathbf{v} + e^{4t} \mathbf{u} = te^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + e^{4t} \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} = e^{4t} \begin{pmatrix} \frac{1}{2} - t \\ 2t \end{pmatrix}$$

These $\mathbf{x}_1$ and $\mathbf{x}_2$ forms a fundamental set of solutions of the system, and we obtain the general solution of $\mathbf{y}' = \mathbf{A}\mathbf{y}$:

$$\mathbf{y} = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t) = c_1 e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} \frac{1}{2} - t \\ 2t \end{pmatrix}.$$

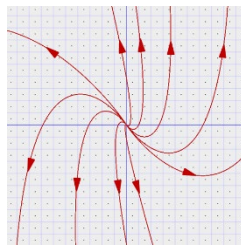
4.7.3c General Solution

If a 2×2 SDE has repeated eigenvalue r with only a single linearly independent eigenvector $\mathbf{v}$, a second linearly independent solution can be created as follows:

- Step 1. Find an eigenvector $\mathbf{v}$ of $\mathbf{A}$ corresponding to eigenvalue r .
- Step 2. Find a nonzero vector $\mathbf{u}$ such that $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$. (There are infinitely many solutions for $\mathbf{u}$. You can choose any particular solution for the next step.)
- Step 3. The two solutions $\mathbf{x}_1 = e^{rt}\mathbf{v}$ and $\mathbf{x}_2 = e^{rt}(\mathbf{tv} + \mathbf{u})$ give a fundamental set of solutions for the system.
- Step 4. Form the general solution $\mathbf{y}(t) = c_1 e^{rt}\mathbf{v} + c_2 e^{rt}(\mathbf{tv} + \mathbf{u})$ of the system.

4.7.3d Trajectories

The phase portrait of the system:



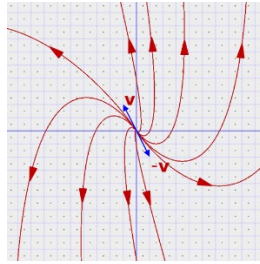
We know the equilibrium point in this case is a degenerate node. As the only eigenvalue is positive, trajectories will emerge from the origin and move away from it, and hence give an unstable equilibrium point.

For both large negative and large positive values of t , the solution will be dominated by the term $t \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ in the general solution:

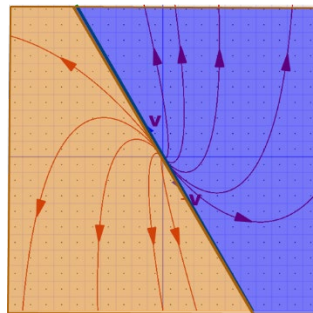
$$\mathbf{y} = c_1 e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} \frac{1}{2} - t \\ 2t \end{pmatrix} = c_1 e^{4t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + c_2 e^{4t} \left[t \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \right]$$

So the sum is almost parallel to the eigenvector $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ for large negative and large positive values of t .

As a result, the trajectories will start in a direction that is parallel to the eigenvector $\mathbf{v}$. They will start in one direction before turning around and moving off into the other direction (parallel to the eigenvector $\mathbf{v}$ again).



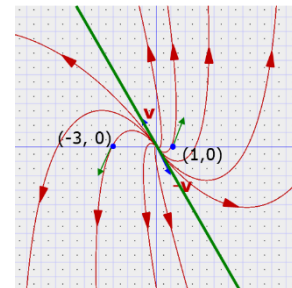
Note that, for trajectories on opposite sides of the eigenvector, the starting directions in which they move are opposite. In this case, all trajectories on the left side of the eigenvector move in the direction of the second quadrant, whereas those on the right side of the eigenvectors move in the direction of the fourth quadrant.



To determine the starting direction of the trajectories, we will take a point on the right side of the eigenvector, say $(1,0)$ and compute the corresponding tangent vector by pre-multiplying the matrix **A**:

$$\begin{pmatrix} 2 & -1 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

Since the tangent vector $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ is pointing upward into first quadrant, it means the trajectories on the right side of eigenvector moves down into the 4th quadrant before turn around into the 1st quadrant.



Although it is not necessary, let us just test another point, this time on the left of the eigenvector, say $(-3, 0)$.

$$\begin{pmatrix} 2 & -1 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} -3 \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ -12 \end{pmatrix}$$

We see that the tangent vector is pointing downward into the 3rd quadrant. This means the trajectories on the left side of eigenvector moves up into the 2nd quadrant before turn around into the 3rd quadrant.